

# RavenStack SaaS Product, Revenue & Churn Analysis

An End-to-End Data Analytics Project using SQL, Python, Statistics and  
Tableau

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Prepared as a beginner-to-intermediate data analytics portfolio project  
Dataset: RavenStack (fully synthetic SaaS dataset) — created by River @  
Rivalytics

*This project uses a synthetic dataset. RavenStack is not a real company and no real customers, accounts, or transactions are represented.*

## 2. Executive Summary

RavenStack is a simulated B2B SaaS company preparing for broader product growth. This project analyzes RavenStack's (synthetic) accounts, subscriptions, product usage, support tickets, and churn events to answer four questions: how customers use the product, which plans and groups generate the most revenue, what factors are associated with churn, and whether product usage and support experience differ between retained and churned customers.

The account base totals **500** accounts generating **\$1,257,903** in current monthly recurring revenue (MRR), with an overall account churn rate of **22.0%** (110 of 500 accounts). Revenue is concentrated in Enterprise-tier accounts, and MRR differs significantly by plan tier. However, several commonly assumed churn drivers — product usage volume and support satisfaction — did **not** show a statistically significant relationship with churn in this dataset. The strongest churn-rate differences observed were across industry and acquisition (referral) channel.

The analysis combines SQL (window functions, CTEs, joins, conditional aggregation), Python-based statistics (descriptive statistics, correlation, hypothesis testing, simple regression), and a two-dashboard Tableau design. All findings are reported using cautious, "associated with" language rather than causal claims, and any non-significant result is reported honestly rather than dressed up as a finding.

## 3. Business Problem

RavenStack is preparing for broader product growth and wants a clearer, data-driven understanding of its customer base ahead of that push. Leadership has four open questions:

- How are customers actually using the product day to day?
- Which plans and customer groups generate the most revenue?
- Which factors are associated with customers churning?
- Do product usage and support experience differ between customers who stay and those who leave?

This project answers each question using only the data available in the five RavenStack tables, without inventing business metrics the data cannot support.

## 4. Dataset Overview

RavenStack is a **fully synthetic** SaaS dataset created by **River @ Rivalytics**, used here for analytics practice. It consists of five related CSV files:

| Table           | Rows   | Description                                         |
|-----------------|--------|-----------------------------------------------------|
| accounts        | 500    | One row per customer account                        |
| subscriptions   | 5,000  | History of subscription/billing periods per account |
| feature_usage   | 25,000 | Daily feature usage events                          |
| support_tickets | 2,000  | Support tickets raised by accounts                  |
| churn_events    | 600    | Log of churn occurrences (including reactivations)  |

All five primary keys (account\_id, subscription\_id, usage\_id, ticket\_id, churn\_event\_id) are unique with zero duplicate rows. Missing values are limited to fields that are legitimately optional: open subscriptions have no end\_date, unanswered CSAT surveys leave satisfaction\_score blank, and churn events without written

feedback leave `feedback_text` blank. Full column definitions are in **DATA\_DICTIONARY.md**.

## 5. Data Relationships

All foreign keys check out cleanly: every `subscriptions.account_id`, `support_tickets.account_id`, and `churn_events.account_id` exists in `accounts`, and every `feature_usage.subscription_id` exists in `subscriptions`.

One important nuance: each account has **~10 subscription records on average** (min 2, max 19), meaning the `subscriptions` table is a history of billing periods, not one row per account. Throughout this project, an account's **"current"** subscription is defined as the record with the most recent `start_date` for that account.

A second nuance worth flagging honestly: `accounts.plan_tier` (an account attribute) frequently disagrees with the plan tier on that account's latest subscription record — the two are essentially independent in this synthetic dataset. To keep the analysis internally consistent, this project always uses the **subscription-level plan tier** when discussing revenue by plan (since MRR is a subscription-level fact), and the `accounts` table for general account attributes like industry and country.

## 6. Data Cleaning

Data cleaning was performed in **Notebook 1** (`01_data_cleaning_and_edu.ipynb`) using only Pandas and NumPy. Steps taken:

- Loaded all 5 CSVs with proper date parsing (`signup_date`, `start_date`, `end_date`, `usage_date`, `submitted_at`, `closed_at`, `churn_date`).
- Checked shape, dtypes, missing values, and duplicate rows for every table (see Section 4).
- Verified primary key uniqueness and foreign key relationships across all 5 tables (see Section 5).
- Created a `duration_days` column for ended subscriptions.
- Rolled feature usage and support tickets up from event-level to account-level summaries (totals, averages, counts).
- Built one denormalized **master account-level table** joining `accounts`, current subscription, usage summary, and support summary — saved to `data/processed/master_account_table.csv`.
- Added simple rule-based Low/Medium/High segments for usage and MRR using quantile thresholds (no clustering algorithms).

The original raw CSV files were never modified — all cleaning outputs are written to `data/processed/`.

## 7. SQL Analysis

SQL was the primary analytical component of this project. **40 queries** are organized across 6 files in `/sql`, covering every required technique:

| File                                 | Queries | Focus                                                        |
|--------------------------------------|---------|--------------------------------------------------------------|
| <code>01_basic_analysis.sql</code>   | 7       | Row counts, breakdowns by industry/country/plan, current MRR |
| <code>02_revenue_analysis.sql</code> | 7       | Revenue by plan/industry, MRR ranking, trend, segmentation   |
| <code>03_product_usage.sql</code>    | 6       | Top features, error rates, beta adoption, usage by plan      |

|                         |   |                                                                   |
|-------------------------|---|-------------------------------------------------------------------|
| 04_churn_analysis.sql   | 7 | Churn rate by segment, churn reasons, usage/support vs churn      |
| 05_support_analysis.sql | 6 | Ticket volume, resolution time, satisfaction, escalations         |
| 06_advanced_sql.sql     | 7 | Extra window functions, 5-table join, query optimization examples |

#### Techniques demonstrated (all validated to run correctly against the dataset):

- **Window functions:** ROW\_NUMBER() to select each account's latest subscription, RANK()/DENSE\_RANK() to rank accounts and plans by MRR, LAG() for month-over-month revenue comparison, and running SUM() OVER() for cumulative MRR.
- **CTEs:** multi-step chains such as aggregating usage per subscription → rolling up to account level → comparing churned vs. retained accounts.
- **Complex joins:** 3+ table joins such as feature\_usage → subscriptions → accounts (for beta feature adoption), and a 5-table join (accounts + subscriptions + usage + tickets + churn) for a denormalized spot-check view.
- **Subqueries:** e.g. finding accounts whose MRR is above the overall average MRR.
- **Conditional aggregation:** SUM(CASE WHEN...) / COUNT(CASE WHEN...) for churned/retained counts, upgrades/downgrades, and ticket-volume buckets.
- **Date functions:** strftime() for monthly grouping, julianday() for subscription duration and days-to-churn.
- **CASE WHEN segmentation:** Low/Medium/High usage, Low/Medium/High MRR, Small/Medium/Large account size.
- **Basic query optimization:** two EXPLAIN QUERY PLAN examples comparing selecting only needed columns and filtering early vs. a naive SELECT \*, plus a note on where indexes would help in production.

## 8. Product Usage Analysis

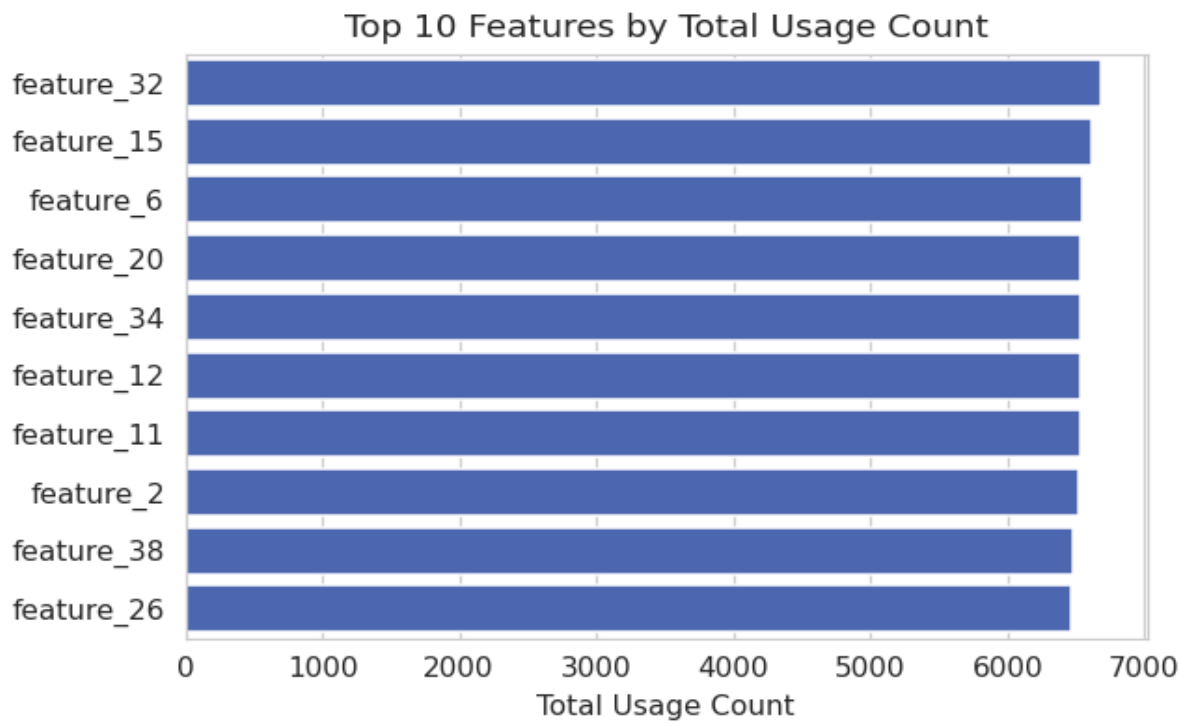


Figure 1. Top 10 features by total usage count.

Feature usage is fairly evenly spread across the catalog, with **feature\_32** the single most-used feature (6,686 total uses). Beta features account for about **10.2%** of all usage events, suggesting reasonable but not dominant beta adoption. Error rates vary meaningfully by feature — a handful of features (feature\_4, feature\_9, feature\_26) show error rates above 0.6 errors per usage event, notably higher than the catalog average, and may be worth a product-quality review.

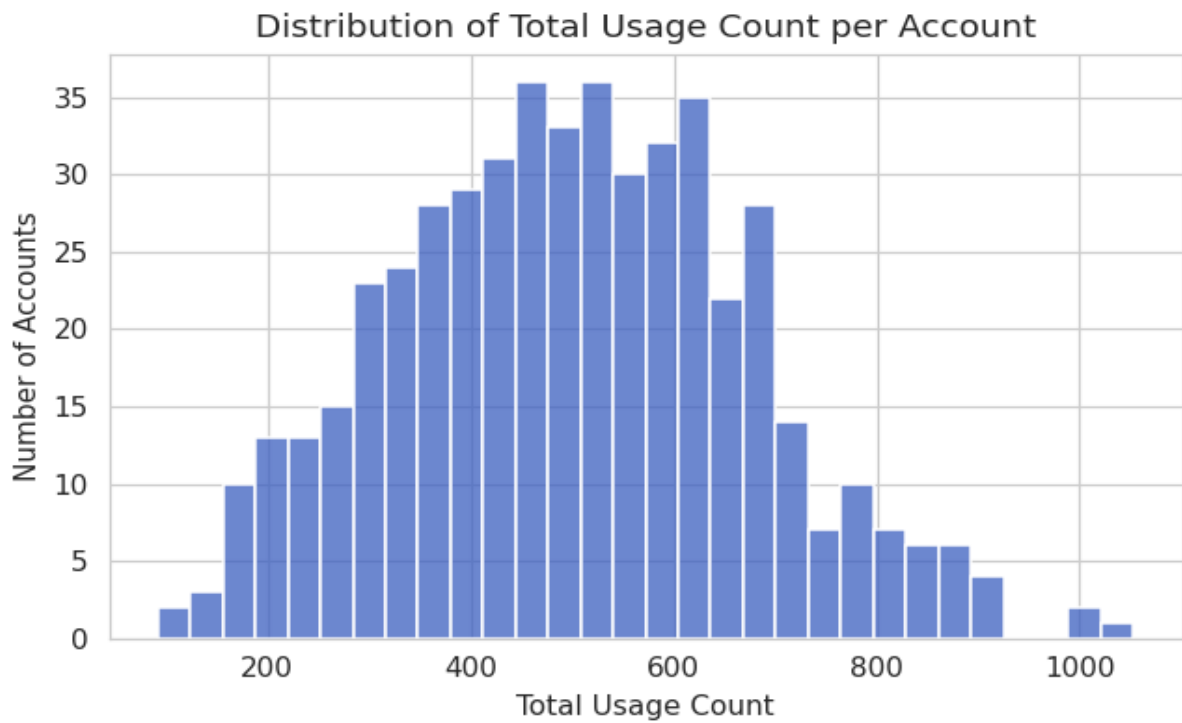


Figure 2. Distribution of total usage count per account.

Usage by plan tier is nearly identical across Basic, Pro, and Enterprise (all around 10 average uses per event) — customers do not appear to use the product noticeably more or less based on which plan they're on.

## 9. Revenue Analysis

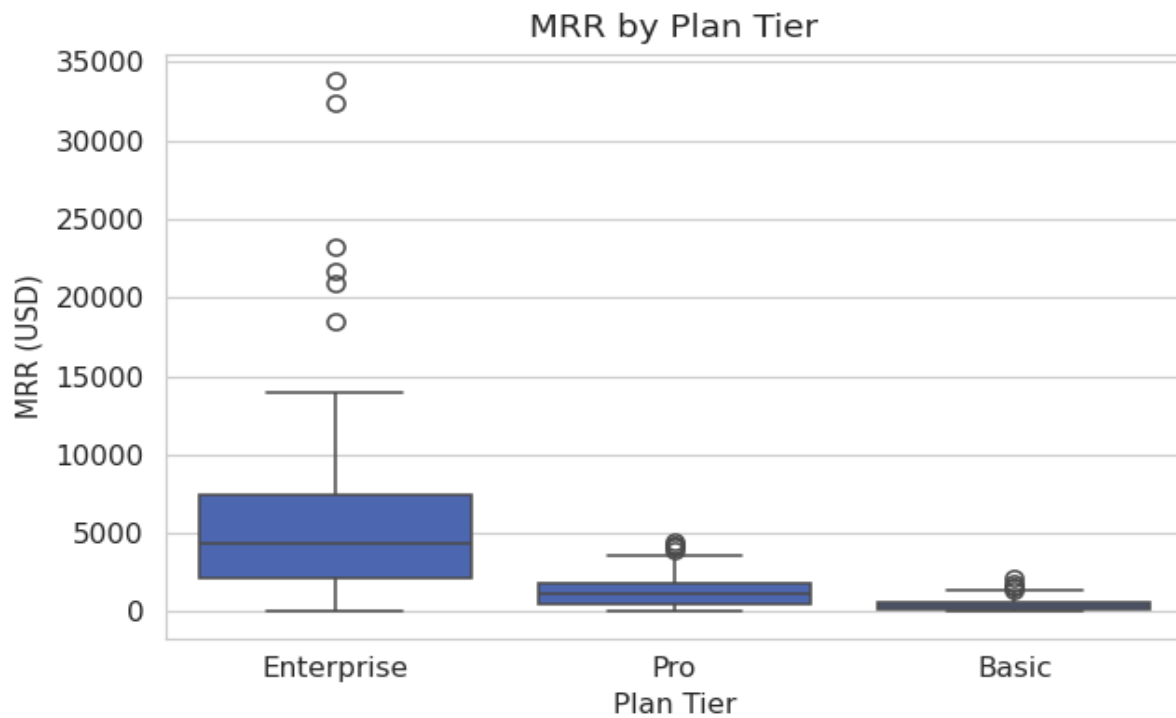


Figure 3. Current MRR distribution by plan tier.

| Plan Tier  | Total MRR | Avg MRR    | Accounts |
|------------|-----------|------------|----------|
| Enterprise | \$969,329 | \$5,507.55 | 176      |
| Pro        | \$214,816 | \$1,317.89 | 163      |
| Basic      | \$73,758  | \$458.12   | 161      |

Enterprise accounts make up roughly 35% of accounts but generate about **77%** of total current MRR (\$969,329 of \$1,257,903) — revenue is heavily concentrated in the top plan tier. An ANOVA test (Section 14) confirms this difference is statistically significant. Average MRR overall is **\$2,515.81**, but the **median** is only **\$1,010.50** — the gap between mean and median confirms a right-skewed distribution driven by a smaller number of high-value accounts.

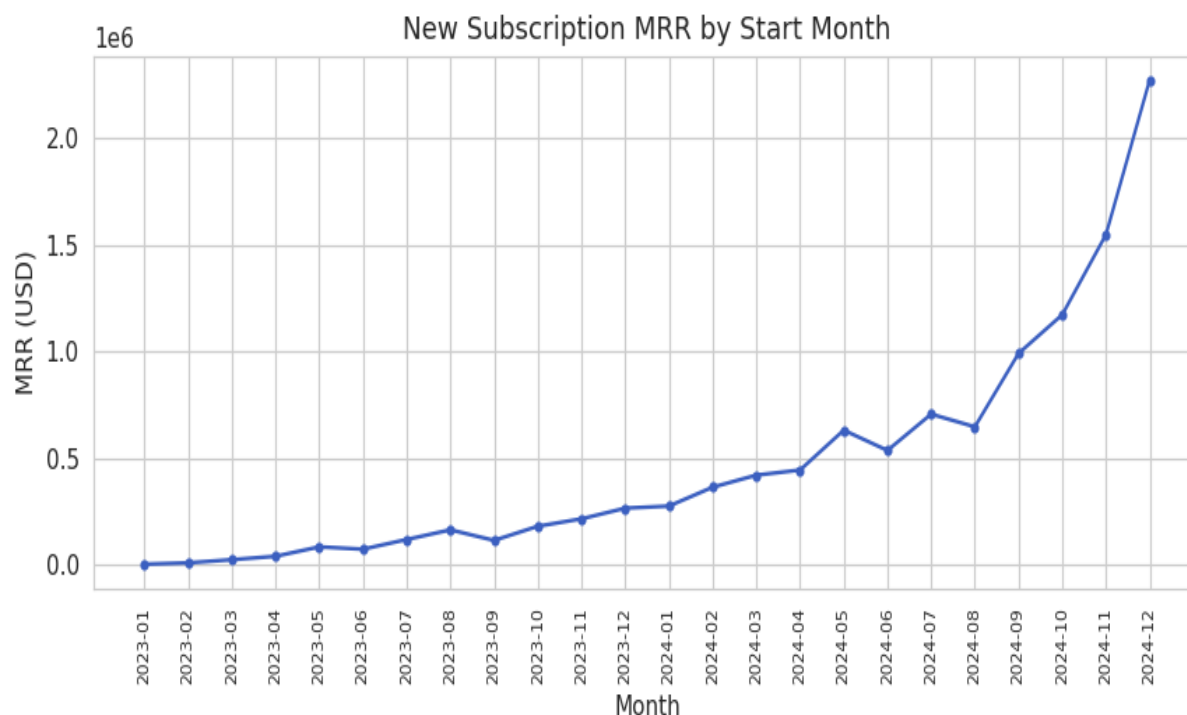


Figure 4. New-subscription MRR booked by month.

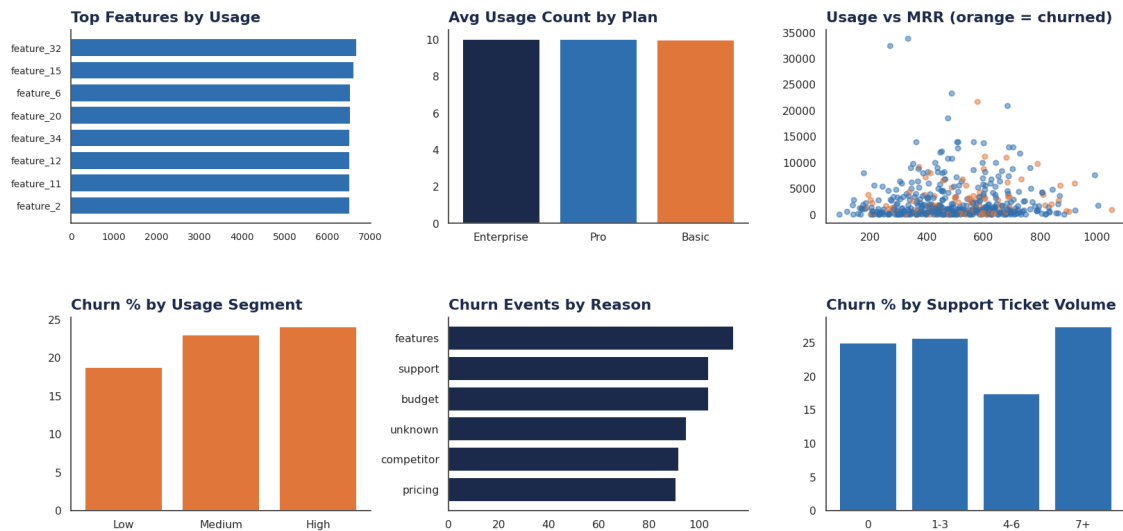
New-subscription MRR booked per month shows a clear upward trend over the observed period, consistent with a growing subscriber base.

## 10. Churn Analysis

The overall account churn rate is **22.0%** (110 of 500 accounts). Average time from signup to churn is approximately **172 days**, and about **10.2%** of churn events are later reactivations.

## Product Usage & Customer Churn

Tableau-style dashboard mockup | Synthetic SaaS dataset (River @ Rivalytics)



Filters (Tableau): Plan | Industry | Referral Source

Figure 5. Product Usage & Customer Churn dashboard mockup.

### 10.1 Churn by segment

| Dimension         | Highest churn segment | Lowest churn segment  | Spread  |
|-------------------|-----------------------|-----------------------|---------|
| Industry          | DevTools (31.0%)      | Cybersecurity (16.0%) | ~15 pts |
| Referral source   | Event (30.2%)         | Partner (14.6%)       | ~16 pts |
| Billing frequency | Annual (24.6%)        | Monthly (18.8%)       | ~6 pts  |
| Plan tier         | Enterprise (23.3%)    | Pro (20.3%)           | ~3 pts  |
| Usage segment     | High (24.1%)          | Low (18.8%)           | ~5 pts  |

Industry and referral source show the widest churn-rate spreads observed in this analysis. Plan tier and usage segment show only modest differences, and are further shown to be statistically non-significant in Section 14. Notably, annual billing shows a slightly *higher* churn rate than monthly — counter to the usual assumption that annual contracts are "stickier" — which may reflect how contract non-renewal is logged at the annual boundary, and is worth further investigation rather than an assumption either way.

### 10.2 Churn reasons

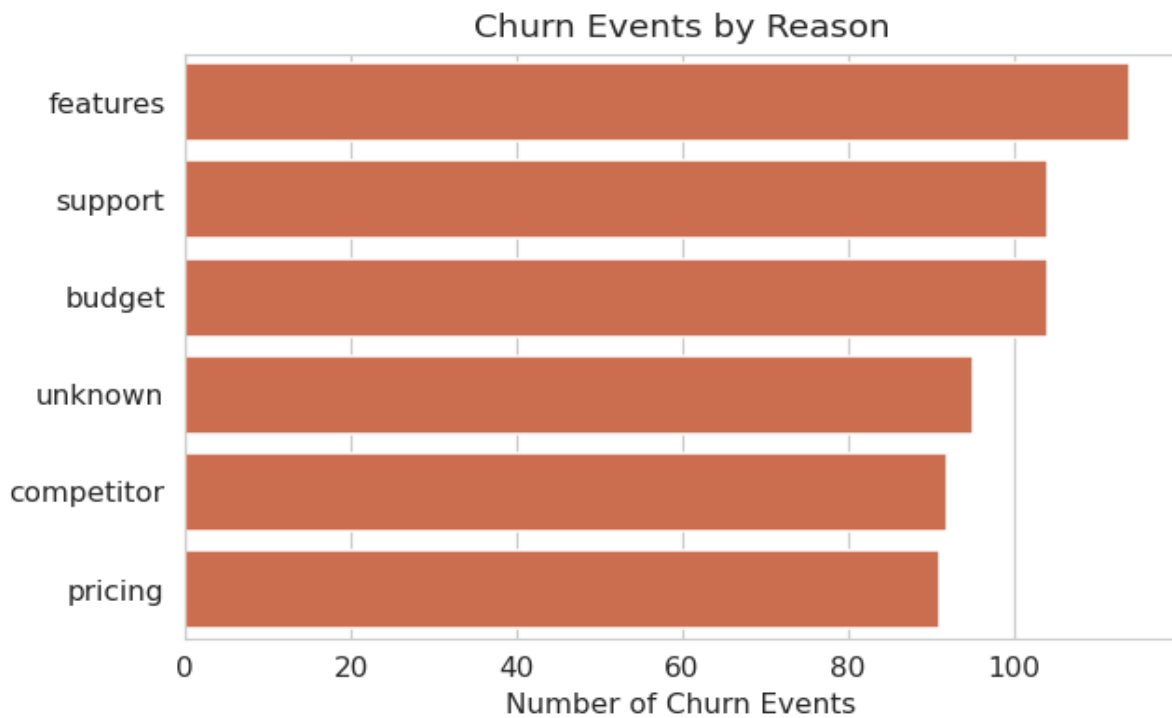


Figure 6. Churn events by reported reason.

Churn reasons are spread fairly evenly across **features** (114 events), **support** (104), and **budget** (104), with competitor and pricing close behind. No single reason dominates, suggesting churn at RavenStack is driven by several factors rather than one fixable root cause.

## 11. Support Analysis

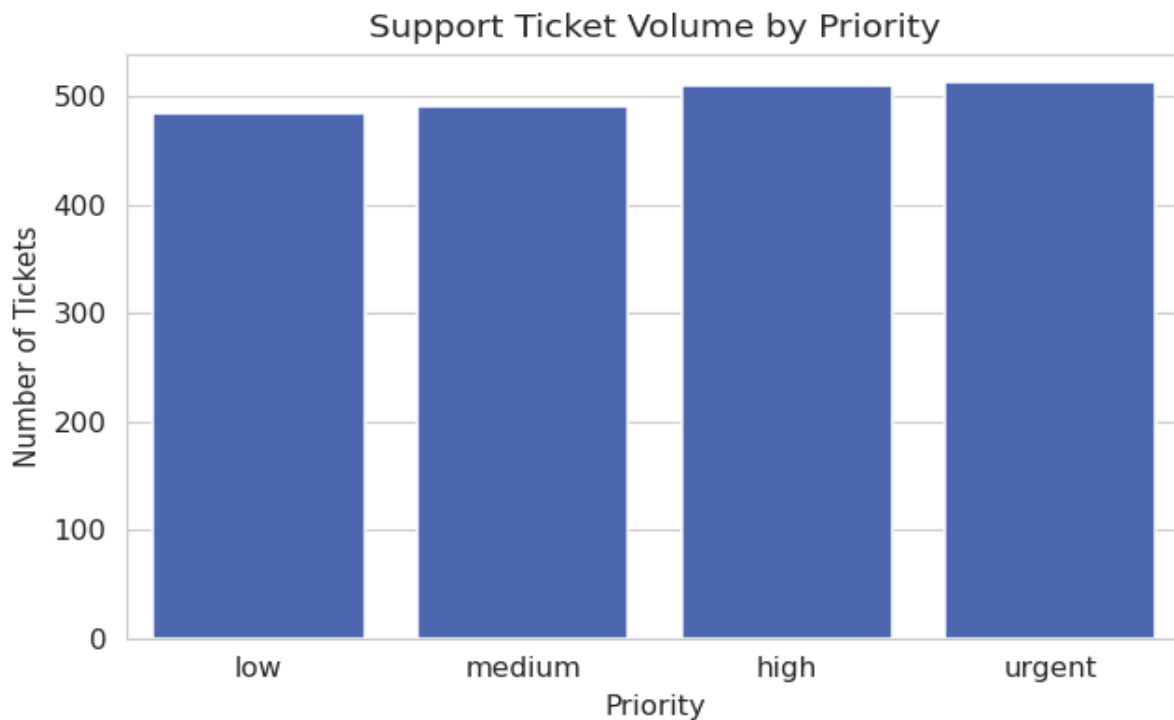


Figure 7. Support ticket volume by priority.

Ticket volume is spread fairly evenly across priority levels (urgent: 514, high: 510, medium: 491, low: 485). The overall escalation rate is low, at **4.75%** of tickets.

### 11.1 Support experience: churned vs. retained

| Metric                    | Retained | Churned |
|---------------------------|----------|---------|
| Avg tickets / account     | 3.95     | 3.93    |
| Avg resolution time (hrs) | 36.45    | 35.49   |
| Avg satisfaction (1-5)    | 3.95     | 4.00    |
| Avg escalations / account | 0.18     | 0.22    |

Support metrics are nearly identical between churned and retained customers. Somewhat counter-intuitively, churned customers report a marginally *higher* average satisfaction score. This is formally tested in Section 14 and found to be statistically non-significant — support experience, at least as captured by these metrics, does not appear to be a standalone churn driver in this dataset.

## 12. Descriptive Statistics

Key numerical variables, compared between retained and churned accounts:

| Variable | Ret. Mean | Ret. Median | Ret. StDev | Churn Mean | Churn Median | Churn StDev |
|----------|-----------|-------------|------------|------------|--------------|-------------|
| MRR (\$) | 2,572     | 992         | 4,072      | 2,318      | 1,152        | 3,166       |
| Seats    | 20.93     | 15.00       | 21.73      | 19.24      | 13.50        | 18.45       |

|              |        |        |        |        |        |        |
|--------------|--------|--------|--------|--------|--------|--------|
| Total Usage  | 495.13 | 491.00 | 172.83 | 522.04 | 527.50 | 176.47 |
| Total Errors | 28.23  | 27.00  | 12.32  | 28.15  | 28.50  | 11.36  |
| Tickets      | 4.02   | 4.00   | 1.84   | 3.93   | 3.50   | 2.08   |
| Satisfaction | 3.95   | 4.00   | 0.57   | 4.00   | 4.00   | 0.56   |

In plain language: churned and retained accounts look very similar on average across usage, errors, tickets, and satisfaction. The most visible difference is in MRR variability — retained accounts show a wider standard deviation, reflecting the mix of small and very large (Enterprise) long-term accounts.

## 13. Correlation Analysis

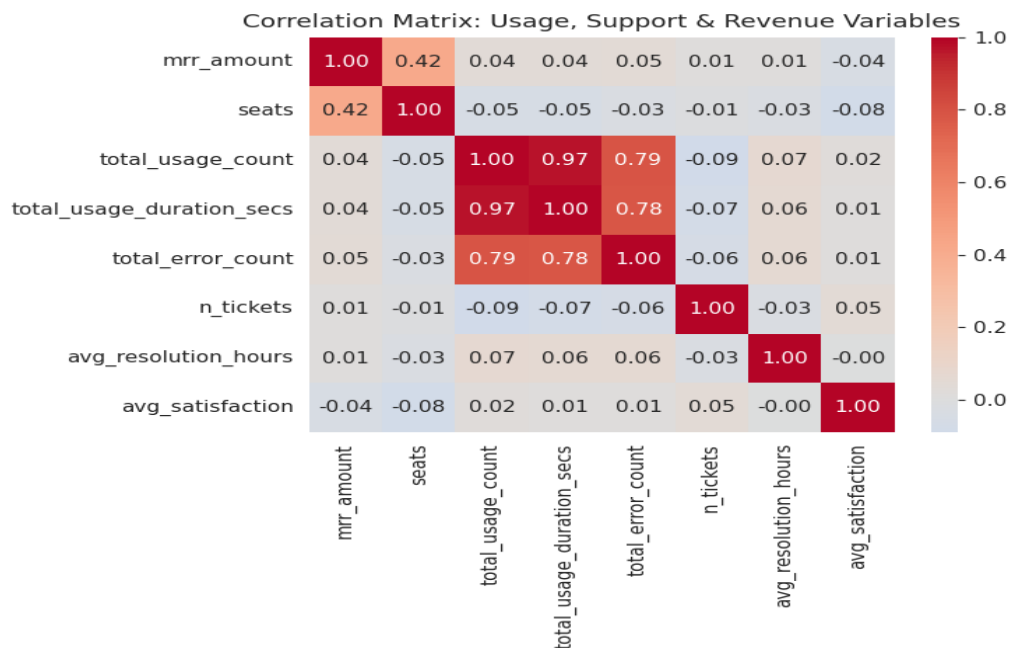


Figure 8. Correlation matrix of usage, support, and revenue variables.

**Direction and strength:** MRR and seats show a moderate positive correlation ( $r \approx 0.43$ ) — larger accounts pay more, as expected under seat-based pricing. Usage variables (total usage count, usage duration, error count) are only weakly correlated with MRR ( $r$  below 0.1). Support variables (tickets, resolution time, satisfaction) show very weak correlations with everything else in the matrix.

**Limitations:** correlation only captures a *linear* relationship between two variables and does **not** imply causation. A weak correlation does not rule out a more complex, non-linear relationship, and a strong correlation could still be driven by a third, unobserved factor. These correlations are used here only to prioritize which relationships were worth testing formally in Section 14.

## 14. Hypothesis Testing

### 14.1 Do churned and retained customers differ in total product usage (total\_usage\_count)?

| H0                                                                                             | H1                                                                                            | Test              | $\alpha$ |
|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-------------------|----------|
| There is no significant difference in mean total usage between churned and retained customers. | There is a significant difference in mean total usage between churned and retained customers. | Two-Sample t-Test | 0.05     |

**Retained mean:** 495.13 **Churned mean:** 522.04

**t-statistic:** -1.419 **p-value:** 0.1578

**Conclusion:**  $p > 0.05 \rightarrow$  **Fail to reject H0.**

**Business meaning:** usage volume alone does not appear to be a reliable early-warning signal for churn in this dataset — RavenStack should not assume more usage automatically means safer from churn.

### 14.2 Do the three plan tiers (Basic, Pro, Enterprise) differ in average MRR?

| H0                                              | H1                                               | Test          | $\alpha$ |
|-------------------------------------------------|--------------------------------------------------|---------------|----------|
| The mean MRR is the same across all plan tiers. | At least one plan tier has a different mean MRR. | One-Way ANOVA | 0.05     |

Group means — Basic: \$458.12, Pro: \$1,317.89, Enterprise: \$5,507.55

**F-statistic:** 122.359 **p-value:** 0.0

**Conclusion:**  $p < 0.05 \rightarrow$  **Reject H0.**

**Business meaning:** pricing tiers are producing meaningfully different revenue outcomes (expected by design), confirming Enterprise accounts are the primary revenue driver.

### 14.3 Do churned customers report lower average support satisfaction scores than retained customers?

| H0                                                                                                      | H1                                                                                                     | Test              | $\alpha$ |
|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-------------------|----------|
| There is no significant difference in mean support satisfaction between churned and retained customers. | There is a significant difference in mean support satisfaction between churned and retained customers. | Two-Sample t-Test | 0.05     |

**Retained mean:** 3.95 **Churned mean:** 4.0

**t-statistic:** -0.779 **p-value:** 0.4373

**Conclusion:**  $p > 0.05 \rightarrow$  **Fail to reject H0.**

**Business meaning:** RavenStack should be cautious about assuming poor support satisfaction is a major, dataset-wide churn driver based on this simple comparison.

## 14.4 Bonus check: Is churn associated with plan tier? (Chi-square test)

| H0                                        | H1                                         | Test                            | $\alpha$ |
|-------------------------------------------|--------------------------------------------|---------------------------------|----------|
| Churn status is independent of plan tier. | Churn status is associated with plan tier. | Chi-square test of independence | 0.05     |

**Chi-square statistic:** 0.477 **dof:** 2 **p-value:** 0.7879

**Conclusion:**  $p > 0.05 \rightarrow$  **Fail to reject H0.** Churn does not appear to be associated with plan tier.

## 15. Hypothetical A/B Test

**Important:** RavenStack's dataset does not contain a real A/B experiment. This section presents a clearly-labeled, **simulated** example of how RavenStack could evaluate a future onboarding experiment. These numbers are illustrative only and are not derived from the real RavenStack dataset used elsewhere in this report.

**Scenario:** RavenStack wants to test whether a new onboarding checklist improves product activation (defined as: did the account use at least 3 different features within its first 14 days?).

- **Control:** existing onboarding flow ( $n = 250$ , simulated)
- **Treatment:** new onboarding checklist ( $n = 250$ , simulated)
- **Primary metric:** activation rate within 14 days
- **H0:** Activation rate is the same for control and treatment.
- **H1:** Activation rate is higher for the treatment group.
- **Test:** two-proportion z-test

| Group                 | n   | Activated | Activation Rate |
|-----------------------|-----|-----------|-----------------|
| Control (simulated)   | 250 | 92        | 36.8%           |
| Treatment (simulated) | 250 | 115       | 46.0%           |

**Simulated result:**  $z \approx 2.16$ , one-sided p-value  $\approx 0.015$ . With  $p < 0.05$ , we would reject H0 in this simulated example and conclude the new onboarding checklist significantly increases activation. **Again, this is a simulated result for teaching purposes** — before acting on a finding like this, RavenStack would need a properly randomized, pre-registered experiment with real assignment and a fixed test duration.

## 16. Regression Analysis

### 16.1 Can customer usage help explain differences in MRR?

**Dependent variable:** mrr\_amount    **Independent variable:** total\_usage\_count ( $n = 500$ )

| Slope  | Intercept | R <sup>2</sup> | p-value | Conclusion                    |
|--------|-----------|----------------|---------|-------------------------------|
| 0.9676 | 2030.97   | 0.0019         | 0.3346  | Not statistically significant |

**Interpretation:**  $R^2 = 0.0019$  means total usage count explains under 1% of the variation in MRR, and the relationship is not statistically significant ( $p = 0.3346 > 0.05$ ). Usage alone is a poor predictor of MRR in this dataset.

## 16.2 A stronger predictor: number of seats

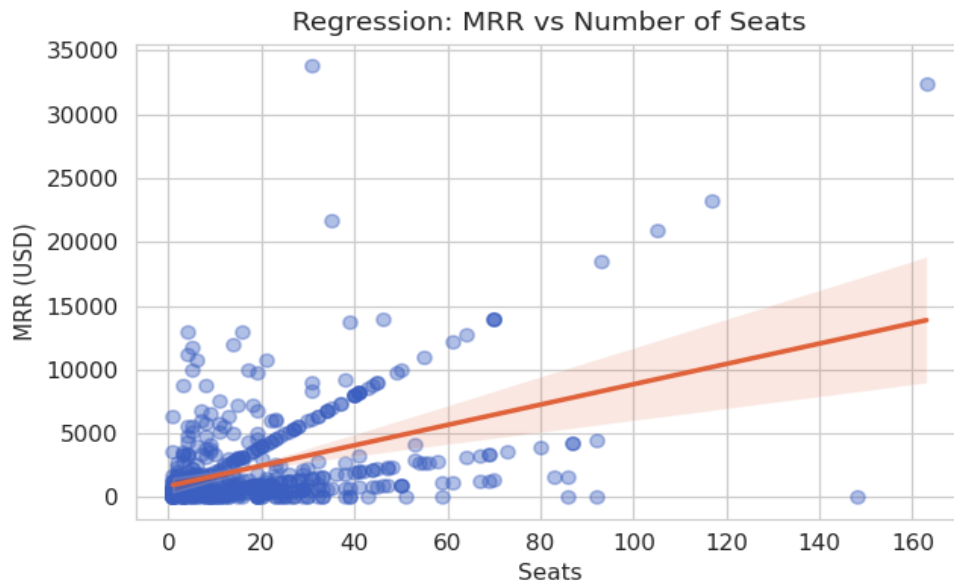


Figure 9. Regression of MRR on number of seats.

| Slope   | Intercept | R <sup>2</sup> | p-value | Conclusion                |
|---------|-----------|----------------|---------|---------------------------|
| 79.7294 | 876.57    | 0.1861         | < 0.001 | Statistically significant |

**Interpretation:** each additional seat is associated with about **\$80** more MRR. Seats explain about **18.6%** of the variance in MRR ( $R^2 = 0.1861$ ), and the relationship is statistically significant ( $p < 0.001$ ) — a far stronger, more intuitive relationship than usage.

**Limitations:** this is a simple one-variable (bivariate) linear regression. It does not control for plan tier, industry, or other factors simultaneously, and an  $R^2$  of  $\sim 0.19$  still leaves most of the variation in MRR unexplained. This is regression *fundamentals* for learning purposes, not a production pricing model.

## 17. Key Findings

- RavenStack's current total MRR is **\$1,257,903** across 500 accounts (avg. \$2,515.81/account); revenue is heavily concentrated in Enterprise accounts ( $\sim 77\%$  of total MRR from  $\sim 35\%$  of accounts).
- The overall account churn rate is **22.0%** (110 of 500 accounts).
- MRR differs significantly across plan tiers (ANOVA,  $p < 0.001$ ) — confirming pricing tiers are functioning as designed.
- Churned and retained customers do **not** differ significantly in total product usage ( $p = 0.158$ ) or average support satisfaction ( $p = 0.437$ ).
- Churn is **not** significantly associated with plan tier (chi-square test,  $p = 0.788$ ).
- The widest churn-rate spreads are by **industry** (DevTools 31.0% vs. Cybersecurity 16.0%) and **referral source** (event-acquired 30.2% vs. partner-referred 14.6%).
- Number of seats is a much stronger, statistically significant predictor of MRR ( $R^2 \approx 0.186$ ,  $p < 0.001$ ) than product usage ( $R^2 \approx 0.002$ , not significant).

- Churn reasons are spread fairly evenly across features, support, and budget — there is no single dominant churn cause.
- Annual billing shows a slightly higher churn rate (24.6%) than monthly billing (18.8%) — counter to the common assumption that annual contracts are stickier, and worth further investigation.
- Beta features account for about 10.2% of all usage events; a handful of features show notably higher error rates and may warrant a product-quality review.

## 18. Recommendations

### 18.1 Recommendation

**Finding:** DevTools accounts churn at 31.0% versus 16.0% in Cybersecurity — the widest industry spread observed.

**Recommendation:** Run a qualitative review (support tickets + churn feedback text) specifically on DevTools accounts to identify whether a product-fit or workflow gap is unique to that segment.

**Expected benefit:** A targeted fix for the highest-churn industry segment could meaningfully move the overall 22.0% churn rate.

### 18.2 Recommendation

**Finding:** Event-sourced accounts churn at 30.2% versus 14.6% for partner referrals — the widest referral-source spread observed.

**Recommendation:** Re-evaluate event-based acquisition: consider stronger post-event onboarding or lead qualification before treating event leads the same as partner referrals.

**Expected benefit:** Improves the quality (retention) of acquired accounts without necessarily changing acquisition volume.

### 18.3 Recommendation

**Finding:** Usage volume and support satisfaction are not statistically significant churn predictors in this dataset.

**Recommendation:** Avoid building a churn early-warning system around usage or satisfaction alone; invest instead in analyzing the free-text churn feedback and industry/channel-level patterns, which showed clearer signal.

**Expected benefit:** Prevents wasted effort on a churn-prediction signal that this data does not support, redirecting analyst time to signals that do show a pattern.

### 18.4 Recommendation

**Finding:** Seats (account size) is a statistically significant, meaningful driver of MRR ( $R^2 \approx 0.19$ ), while usage is not.

**Recommendation:** Prioritize seat-expansion motions (upsell more seats to existing accounts) over usage-based upsell messaging, since seat count is the more proven revenue lever in this data.

**Expected benefit:** Focuses expansion revenue efforts on the lever with demonstrated statistical support.

## 18.5 Recommendation

**Finding:** Enterprise accounts generate ~77% of total MRR from ~35% of accounts, but do not show a lower churn rate than other tiers (23.3% vs. 20.3% for Pro).

**Recommendation:** Give Enterprise accounts a dedicated retention/success motion, since losing even a small number of these accounts has an outsized revenue impact relative to their churn-rate share.

**Expected benefit:** Protects the revenue base that matters most, rather than spreading retention effort evenly across all tiers.

## 18.6 Recommendation

**Finding:** Churn reasons are evenly split across features, support, and budget, with no single dominant cause.

**Recommendation:** Treat churn as multi-causal in messaging to leadership: pursue three parallel, modest initiatives (a product-feedback loop, a support-quality check, and a pricing/value review) rather than one large initiative aimed at a single root cause.

**Expected benefit:** Matches the intervention strategy to the actual (spread-out) shape of the churn-reason data.

# 19. Tableau Dashboards

Two Tableau-style dashboard mockups were built using the actual calculated numbers from this analysis (no invented KPI values). Full build instructions to recreate these as live, interactive Tableau workbooks are provided in `TABLEAU_BUILD_GUIDE.md`.

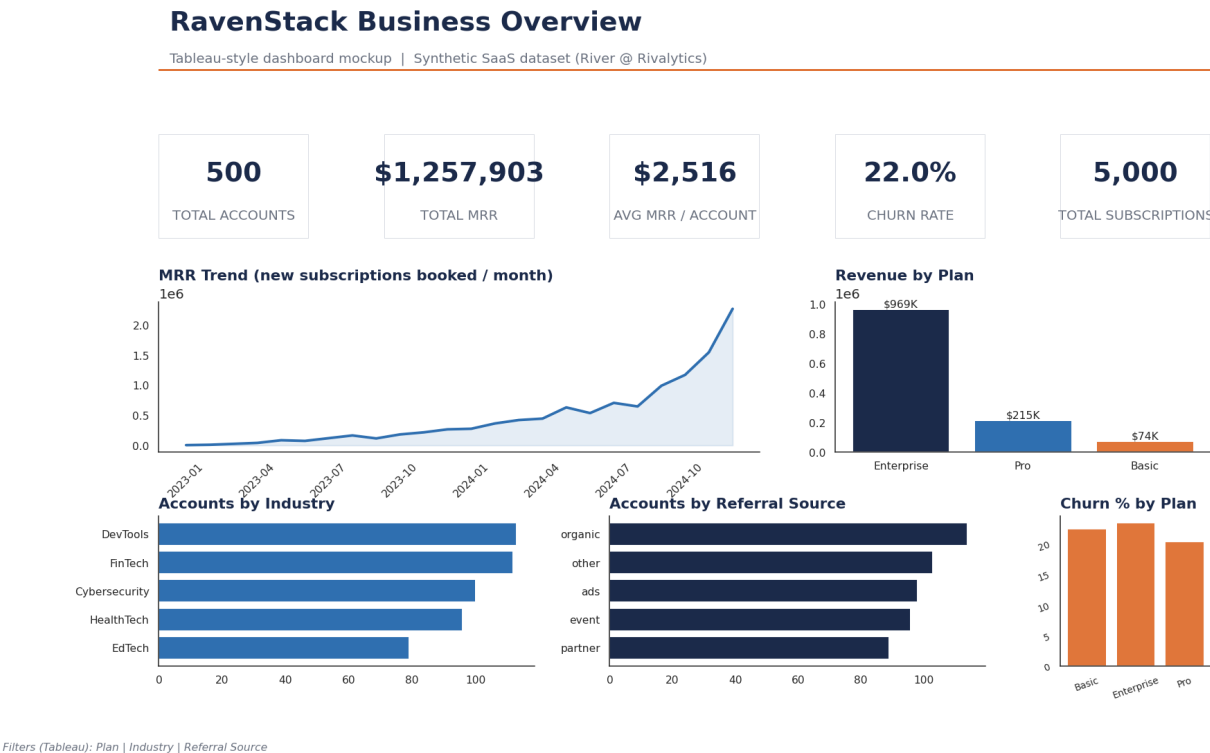
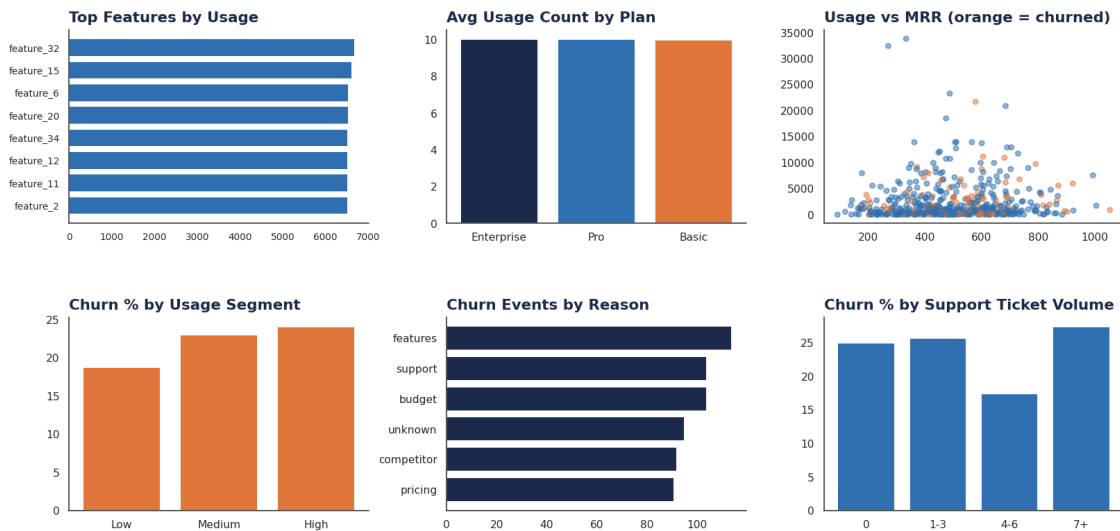


Figure 10. Dashboard 1 — RavenStack Business Overview.

## Product Usage & Customer Churn

Tableau-style dashboard mockup | Synthetic SaaS dataset (River @ Rivalytics)



Filters (Tableau): Plan | Industry | Referral Source

Figure 11. Dashboard 2 — Product Usage & Customer Churn.

Note: these are Tableau-style dashboard mockups based on the project's analysis, not screenshots taken from Tableau Desktop.

## 20. Limitations

- This is a **synthetic** dataset; patterns and relationships may not reflect a real SaaS business, and results should not be generalized beyond this exercise.
- accounts.plan\_tier sometimes disagrees with the plan tier on an account's latest subscription — documented and worked around by consistently using subscription-level plan data for revenue analysis.
- Several relationships tested (usage vs. churn, satisfaction vs. churn, plan tier vs. churn) were not statistically significant — reported honestly rather than reframed as findings.
- The A/B test section is a hypothetical, simulated example only; RavenStack's data does not contain a real experiment.
- Regression analysis here is simple bivariate regression for learning purposes, not a multi-variable production model.
- Tableau dashboards are static mockup images built from the calculated numbers, not live interactive workbooks (a build guide is provided to recreate them).

## 21. Conclusion

This project demonstrates a complete, honest analytics workflow on a synthetic SaaS dataset: from data understanding and cleaning, through SQL and Python-based statistical analysis, to dashboard design and business recommendations. The most valuable outcome is not any single number, but the discipline of testing assumptions rather than accepting them — several intuitive "usage drives retention" and "satisfaction drives retention" stories did not hold up under a simple hypothesis test, while less obvious signals (industry, referral channel, seat count) showed clearer, statistically supported patterns. That distinction is exactly the kind of finding a real stakeholder needs before investing in a churn-prevention or revenue-growth initiative.